

Tutorial – 05

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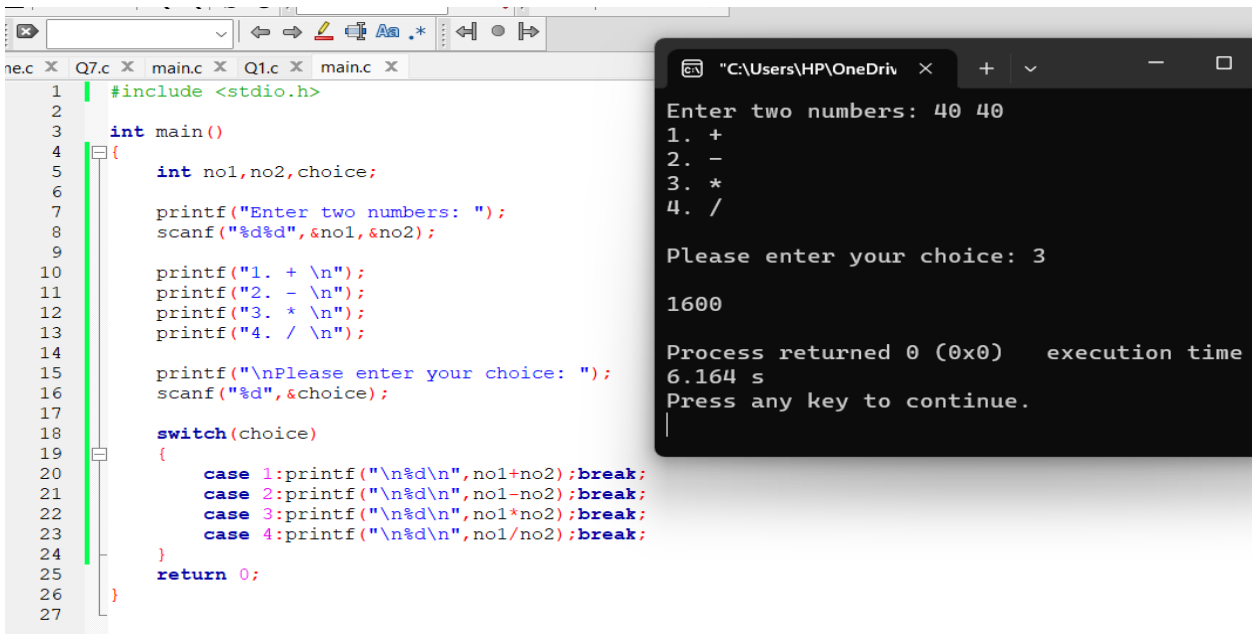
Switch

Input two numbers and display the outputs of the basic mathematic operations. The output screen should be displayed as follows;

Enter two numbers ____

1. +
2. -
3. *
4. /

Please enter your Choice ____



```
1 | #include <stdio.h>
2 |
3 | int main()
4 | {
5 |     int no1, no2, choice;
6 |
7 |     printf("Enter two numbers: ");
8 |     scanf("%d%d", &no1, &no2);
9 |
10 |    printf("1. + \n");
11 |    printf("2. - \n");
12 |    printf("3. * \n");
13 |    printf("4. / \n");
14 |
15 |    printf("\nPlease enter your choice: ");
16 |    scanf("%d", &choice);
17 |
18 |    switch(choice)
19 |    {
20 |        case 1: printf("\n%d\n", no1+no2); break;
21 |        case 2: printf("\n%d\n", no1-no2); break;
22 |        case 3: printf("\n%d\n", no1*no2); break;
23 |        case 4: printf("\n%d\n", no1/no2); break;
24 |    }
25 |    return 0;
26 | }
27 |
```

```
"C:\Users\HP\OneDrive" x + - □
Enter two numbers: 40 40
1. +
2. -
3. *
4. /

Please enter your choice: 3

1600

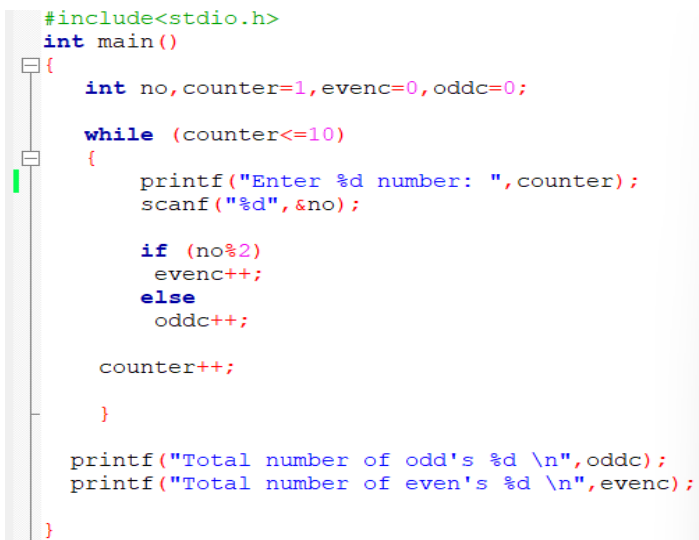
Process returned 0 (0x0)   execution time
6.164 s
Press any key to continue.
```

While loop

1. **Input 10 numbers and display the total count of odd & even numbers in the entered number series.**

```
#include<stdio.h>

int main()
{
    int no,counter=1,evenc=0,oddc=0;
    while (counter<=10)
    {
        printf("Enter %d number: ",counter);
        scanf("%d",&no);
        if (no%2)
            evenc++;
        else
            oddc++;
        counter++;
    }
    printf("Total number of odd's %d \n",oddc);
    printf("Total number of even's %d \n",evenc);
}
```



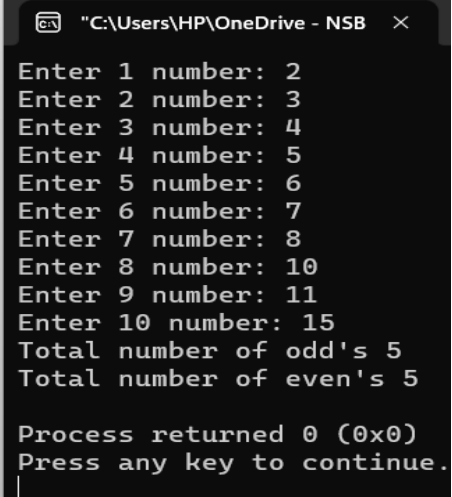
```
#include<stdio.h>
int main()
{
    int no,counter=1,evenc=0,oddc=0;

    while (counter<=10)
    {
        printf("Enter %d number: ",counter);
        scanf("%d",&no);

        if (no%2)
            evenc++;
        else
            oddc++;

        counter++;
    }

    printf("Total number of odd's %d \n",oddc);
    printf("Total number of even's %d \n",evenc);
}
```



```
"C:\Users\HP\OneDrive - NSB"
Enter 1 number: 2
Enter 2 number: 3
Enter 3 number: 4
Enter 4 number: 5
Enter 5 number: 6
Enter 6 number: 7
Enter 7 number: 8
Enter 8 number: 10
Enter 9 number: 11
Enter 10 number: 15
Total number of odd's 5
Total number of even's 5

Process returned 0 (0x0)
Press any key to continue.
|
```

2.Modify the above program in to enter series of numbers terminates when the user enter -99 and display the same expected output.

```
#include<stdio.h>
int main()
{
int no,evenc=0,oddc=0;
printf("Enter numbers (-99 to terminate)\n");
while (1)
{
printf("Number: ");
scanf("%d", &no);
if (no == -99)
{
break;
}
if (no%2)
evenc++;
else
oddc++;
}
printf("Total number of odd's %d \n",oddc);
printf("Total number of even's %d \n",evenc);

}
```

```

x ooo.c x while2.c x
#include<stdio.h>
int main()
{
    int no,evenc=0,oddc=0;

    printf("Enter numbers (-99 to terminate)\n");

    while (1)
    {
        printf("Number: ");
        scanf("%d", &no);

        if (no == -99)
        {
            break;
        }

        if (no%2)
            evenc++;
        else
            oddc++;
    }

    printf("Total number of odd's %d \n",oddc);
    printf("Total number of even's %d \n",evenc);
}

```

Enter numbers (-99 to terminate)

Number: 1

Number: 2

Number: 3

Number: 4

Number: 5

Number: 6

Number: 7

Number: 8

Number: 9

Number: 0

Number: 10

Number: 11

Number: 23

Number: 45

Number: -99

Total number of odd's 6

Total number of even's 8

Process returned 0 (0x0) execution time: 18.189 s

Press any key to continue.

Do while loop

Rewrite the programs for the above while loop question 1 & 2 using do while loop

```
1 #include<stdio.h>
   int main()
   {
       int no,counter=1,oddc=0,evenc=0;
       do
       {
           printf("Enter %d number : ",counter);
           scanf("%d",&no);

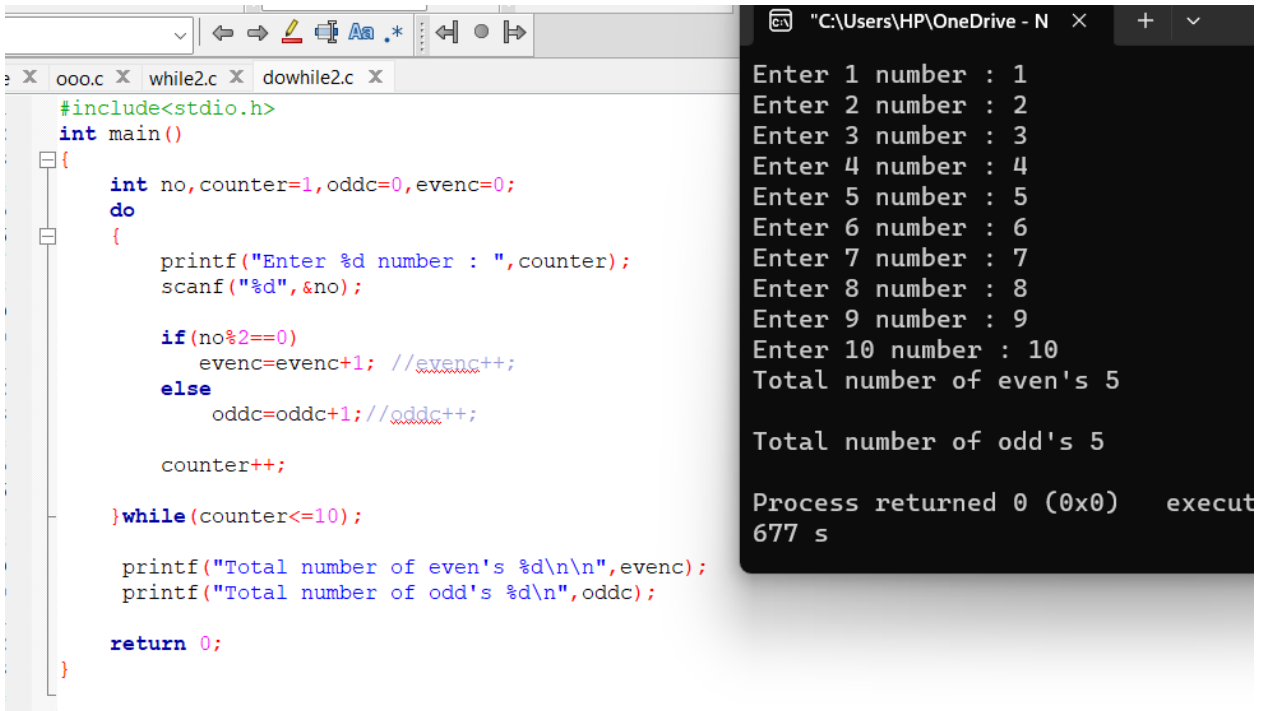
           if(no%2==0)
               evenc=evenc+1; //evenc++;
           else
               oddc=oddc+1;//oddc++;

           counter++;

       }while(counter<=10);

       printf("Total number of even's %d\n\n",evenc);
       printf("Total number of odd's %d\n",oddc);

       return 0;
   }
```

The image shows a code editor window on the left and a terminal window on the right. The code editor displays a C program that uses a do-while loop to count even and odd numbers from 1 to 10. The terminal window shows the program's execution, with prompts for numbers 1 through 10, followed by the total counts for even (5) and odd (5) numbers, and a final message indicating the process returned 0.

```
ooo.c x while2.c x dowhile2.c x
#include<stdio.h>
int main()
{
    int no, counter=1, oddc=0, evenc=0;
    do
    {
        printf("Enter %d number : ", counter);
        scanf("%d", &no);

        if(no%2==0)
            evenc=evenc+1; //evenc++;
        else
            oddc=oddc+1; //oddc++;

        counter++;
    } while(counter<=10);

    printf("Total number of even's %d\n\n", evenc);
    printf("Total number of odd's %d\n", oddc);

    return 0;
}
```

```
Enter 1 number : 1
Enter 2 number : 2
Enter 3 number : 3
Enter 4 number : 4
Enter 5 number : 5
Enter 6 number : 6
Enter 7 number : 7
Enter 8 number : 8
Enter 9 number : 9
Enter 10 number : 10
Total number of even's 5

Total number of odd's 5

Process returned 0 (0x0)   execut
677 s
```

2.

```
#include<stdio.h>
int main()
{
    int no, oddc=0, evenc=0;

    printf("Enter numbers (-99 to terminate)\n");

    do
    {
        printf("Number:");
        scanf("%d", &no);

        if (no == -99)
        {
            break;
        }
    }
}
```

```

    }

    if(no%2==0)
        evenc=evenc+1; //evenc++;
    else
        oddc=oddc+1;//oddc++;

}while(1);

printf("Total number of even's %d\n\n",evenc);
printf("Total number of odd's %d\n",oddc);

return 0;
}

```

```

re x ooo.c x while2.c x dowhile2.c x
1  #include<stdio.h>
2  int main()
3  {
4      int no,oddc=0,evenc=0;
5
6      printf("Enter numbers (-99 to terminate)\n");
7
8      do
9      {
10         printf("Number:");
11         scanf("%d",&no);
12
13         if (no == -99)
14         {
15             break;
16         }
17
18         if(no%2==0)
19             evenc=evenc+1; //evenc++;
20         else
21             oddc=oddc+1;//oddc++;
22     }while(1);
23
24     printf("Total number of even's %d\n\n",evenc);
25     printf("Total number of odd's %d\n",oddc);
26
27     return 0;
28 }
29
30

```

```

"C:\Users\HP\OneDrive - N x + v
Enter numbers (-99 to terminat
Number:2
Number:3
Number:4
Number:5
Number:6
Number:7
Number:8
Number:9
Number:29
Number:30
Number:99
Number:100
Number:-99
Total number of even's 6
Total number of odd's 6
Process returned 0 (0x0) exe
348 s
Press any key to continue.
|

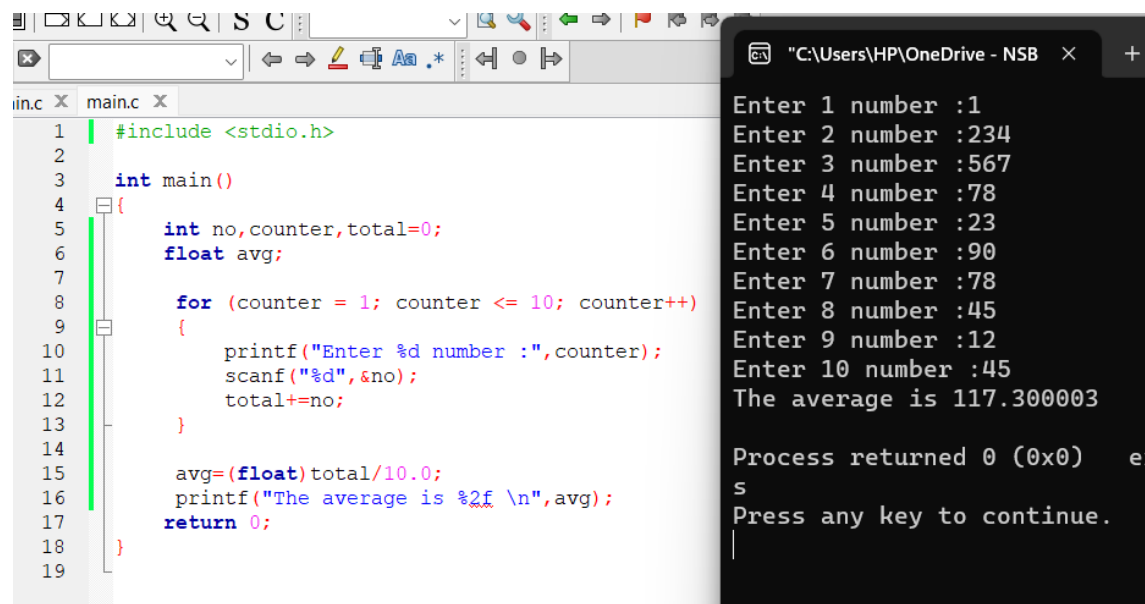
```

For loop

1. Input 10 numbers and display the average value using the for loop

```
#include <stdio.h>

int main()
{
    int no,counter,total=0;
    float avg;
    for (counter = 1; counter <= 10; counter++)
    {
        printf("Enter %d number :",counter);
        scanf("%d",&no);
        total+=no;
    }
    avg=(float)total/10.0;
    printf("The average is %2f \n",avg);
    return 0;
}
```



The screenshot shows a C program in a code editor and its execution in a terminal window. The code calculates the average of 10 numbers. The terminal output shows the input of 10 numbers and the resulting average of 117.300003.

```
inc X main.c X
1 | #include <stdio.h>
2 |
3 | int main()
4 | {
5 |     int no,counter,total=0;
6 |     float avg;
7 |
8 |     for (counter = 1; counter <= 10; counter++)
9 |     {
10 |         printf("Enter %d number :",counter);
11 |         scanf("%d",&no);
12 |         total+=no;
13 |     }
14 |
15 |     avg=(float)total/10.0;
16 |     printf("The average is %2f \n",avg);
17 |     return 0;
18 | }
19 |

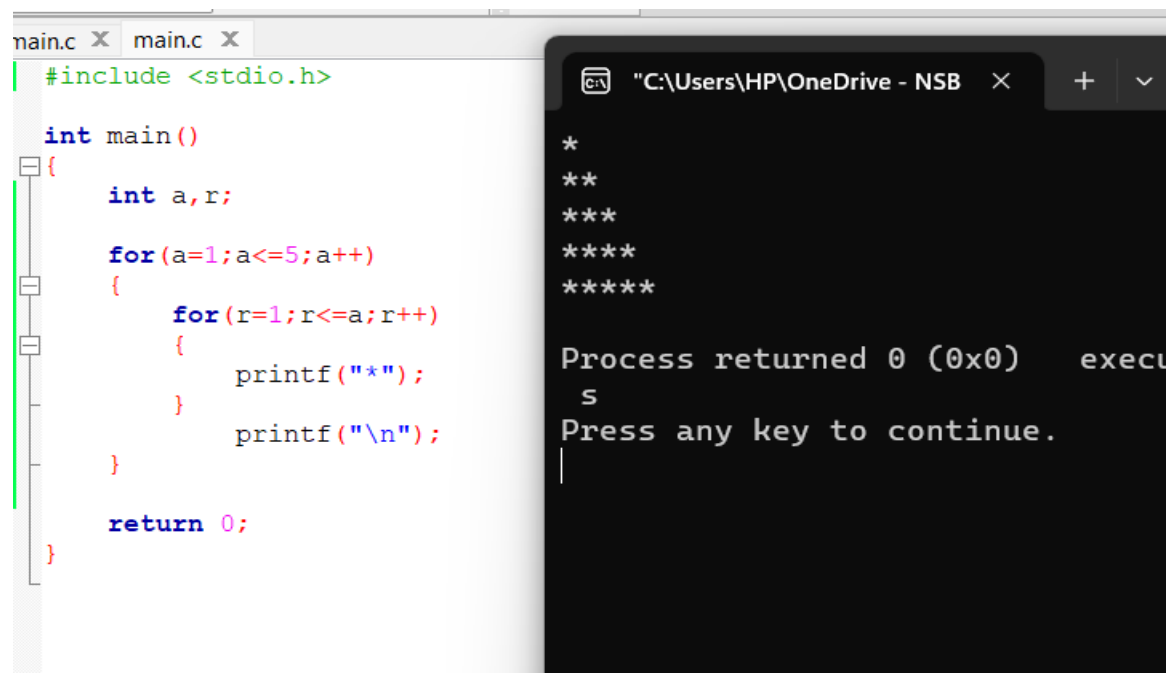
"C:\Users\HP\OneDrive - NSB" x +
Enter 1 number :1
Enter 2 number :234
Enter 3 number :567
Enter 4 number :78
Enter 5 number :23
Enter 6 number :90
Enter 7 number :78
Enter 8 number :45
Enter 9 number :12
Enter 10 number :45
The average is 117.300003

Process returned 0 (0x0)   e
s
Press any key to continue.
|
```


2. Display the following output using the for loop

```
*  
**  
***  
****  
*****
```

```
#include <stdio.h>  
int main()  
{  
    int a,r;  
    for(a=1;a<=5;a++)  
    {  
        for(r=1;r<=a;r++)  
        {  
            printf("*");  
        }  
        printf("\n");  
    }  
    return 0;  
}
```



```
main.c X main.c X  
#include <stdio.h>  
  
int main()  
{  
    int a,r;  
  
    for(a=1;a<=5;a++)  
    {  
        for(r=1;r<=a;r++)  
        {  
            printf("*");  
        }  
        printf("\n");  
    }  
  
    return 0;  
}
```

```
"C:\Users\HP\OneDrive - NSB X + v  
*  
**  
***  
****  
*****  
  
Process returned 0 (0x0)   execu  
s  
Press any key to continue.  
|
```